Swisher's (H.3): Ordinary and Supersingular Primes

A Unified Proof Manuscript with the Small-Prime Correction

Research manuscript

Unified edition: 10 November 2026

Abstract

We present a unified proof of the ordinary and supersingular branches of Swisher's conjecture (H.3) for $p > 3$, together with the corrected supersingular statement at $p = 3$. The common input is an integral Hecke trace in the raw cubic parameter. An explicit extra normalizer makes this parameter a Hauptmodul and gives an all-level formal congruence with precision p^{3s} for every odd prime. At ordinary primes, canonical CM specialization gives the multiplier $-\Gamma_p(1/4)^4$, identified exactly using Guo's weaker published recurrence; arbitrary raw lengths and the full cutoff are also covered. At supersingular primes, the value and first two derivatives of a Legendre pullback are transported to the full integral crystal. An exact differential residual and a weighted Taylor estimate give the two-step modulus p^{3s-1} for $p \geq 7$ and 3^{3s-2} for $p = 3$. The common trace argument is included once, both specialization arguments are included in full, and independent finite identities and reproducible verification programs are supplied as appendices.

Status. This is a newly written research proof manuscript. The arguments have undergone internal mathematical and exact computational checks. External expert review and a definitive assessment of novelty remain pending. Standard modular and crystalline inputs are cited explicitly; the new connecting arguments are written out below.

Contents

1 The combined statement and its precise scope **2**

2 The raw cubic parameter and the half-length truncation **3**

3 The integral Hecke trace at every raw cutoff **4**

4 Integral specialization on the ordinary modular curve **6**

5 The ordinary gamma multiplier and arbitrary-length recurrence **8**

6 Integral Legendre jets and change of Frobenius lift **8**

7 Supersingular CM specialization **12**

8 Precision, scope, and exact computational checks **14**

A An independent finite identity and valuation bound **15**

B Reproducible verification programs **16**

1 The combined statement and its precise scope

For an odd prime p and integers $m \geq 1$, $s \geq 0$, define

$$A_{m,s} = \sum_{0 \leq 2k < mp^s} \frac{(1/2)_k^3}{k!^3}, \qquad H_s = A_{1,s}, \qquad H_0 = 1, \qquad \epsilon_p = \begin{cases} 1, & p = 3, \\ 0, & p \geq 5. \end{cases}$$

The prime is suppressed in this notation; $H_s(3)$ will indicate the specialization $p = 3$ when needed. Here $(a)_k$ is the rising factorial, $v_p(p) = 1$, and Γ_p denotes Morita's p-adic gamma function, the continuous extension to $\mathbb{Z}_p$ of

$$\Gamma_p(n) = (-1)^n \prod_{\substack{1 \leq j < n \\ p \nmid j}} j \qquad (n \geq 1).$$

Theorem 1.1 (Ordinary and supersingular recurrences)**.** *Let p be an odd prime.*

1. *If $p \equiv 1 \pmod 4$, then for every $m, s \geq 1$,*

$$A_{m,s} \equiv -\Gamma_p(1/4)^4 A_{m,s-1} \pmod{p^{3s+3v_p(m)}}. \tag{1}$$

2. *If $p \equiv 3 \pmod 4$, then $H_1 \equiv 0 \pmod{p^2}$ and, for every $s \geq 2$,*

$$H_s \equiv p^2 H_{s-2} \pmod{p^{3s-1-\epsilon_p}}. \tag{2}$$

For $p > 3$, the half-length case $m = 1$ of the first assertion, the supersingular first level, and the supersingular two-step recurrence are the three statements listed as (H.3) in [1, Table 2, (H.3)]. The theorem treats every level in their stated ranges. Its $p = 3$ conclusion uses the necessary correction discussed below.

Prime range	Sum and levels	Recurrence	Modulus
$p \equiv 1 \pmod 4$	$A_{m,s}$, $m, s \geq 1$	one step	$p^{3s+3v_p(m)}$
$p \equiv 3 \pmod 4$	H_1	$H_1 \equiv 0$	p^2
$p \geq 7$, $p \equiv 3 \pmod 4$	H_s, $s \geq 2$	two steps	p^{3s-1}
$p = 3$	H_s, $s \geq 2$	two steps	3^{3s-2}

The ordinary full-cutoff case is $m = 2$:

$$\sum_{k=0}^{p^s-1} \frac{(1/2)_k^3}{k!^3} \equiv -\Gamma_p(1/4)^4 \sum_{k=0}^{p^{s-1}-1} \frac{(1/2)_k^3}{k!^3} \pmod{p^{3s}}.$$

The arbitrary-length and full-cutoff extensions in this manuscript concern ordinary primes only. The supersingular assertion concerns the exact half cutoff $(p^s - 1)/2$.

1.1 The necessary correction at three

The unqualified exponent $3s - 1$ fails at $p = 3$, already at $s = 2$:

$$H_2(3) = \frac{2576763}{2097152}, \qquad H_2(3) - 9 = -\frac{3^4 \cdot 201205}{2^{21}}.$$

Thus $v_3(H_2(3) - 9) = 4$, whereas $3s - 1 = 5$. The exponent in (2) is $3s - 2$ at 3, and is attained by this example. This is a correction to the small-prime statement, not an unresolved exception in the proof of the corrected theorem.

1.2 Common input and the two specialization arguments

Sections 2–3 establish the common formal congruence directly. Retaining the raw parameter makes its p^s truncation exactly the half-length sum in (H.3). The argument uses the modular framework of [3, 5], with the excellent-lift viewpoint of [4], and includes its own Hecke trace induction. No result from a separate unpublished manuscript is assumed.

Sections 4–5 treat ordinary primes. The point $u = 1/64$ is a zero of the second Hasse–Witt factor in the BV3 cubic chart, so specialization is performed on an ordinary modular-curve chart. Guo proved the half- and full-cutoff recurrences modulo p^{s+1} in [2], equations (1.9)–(1.10) in the arXiv version. Only the half-cutoff comparison is used to identify the constant multiplier; the higher precision comes from the trace.

Sections 6–7 start afresh from the all-odd-prime formal theorem on the Legendre base. They do not specialize an ordinary canonical lift at a supersingular point. An integral rank-three jet congruence is first transported to a regular lift on the full crystal, and only then evaluated at $\lambda = 1/2$. This proof does not depend on the ordinary CM or gamma-multiplier conclusion.

2 The raw cubic parameter and the half-length truncation

Put

$$\mathcal{F}(X) = \sum_{k\ge 0} \binom{2k}{k}^3 X^{2k}, \qquad \mathcal{F}_L(X) = \sum_{0\le 2k<L} \binom{2k}{k}^3 X^{2k}.$$

Thus the sum in Swisher's conjecture (H.3) is exactly

$$H_s = \mathcal{F}_{p^s}(1/8) = \sum_{k=0}^{(p^s-1)/2} \frac{(1/2)_k^3}{k!^3}.$$

This identity does not discard a tail of a longer truncation. For $q = e^{2\pi i\tau}$ define

$$x(\tau) = \frac{\eta(2\tau)^{12}\eta(8\tau)^{12}}{\eta(4\tau)^{24}}, \qquad \Theta(\tau) = \frac{\eta(4\tau)^{20}}{\eta(2\tau)^8\eta(8\tau)^8}.$$

The eta products and theta identities used here are classical; see [3, Appendix C, Section 9.1] and [10, Section 20.7]. The product expansions give $x \in q + q^2\mathbb{Z}[[q]]$ and $\Theta \in 1 + q\mathbb{Z}[[q]]$. If $z = 4\tau$, the usual theta identities give

$$x = \frac{\vartheta_2(z)^2\vartheta_4(z)^2}{4\vartheta_3(z)^4}, \qquad \Theta = \vartheta_3(z)^4. \tag{3}$$

Writing $\lambda(z) = \vartheta_2(z)^4/\vartheta_3(z)^4$, one has

$$16x^2 = \lambda(1-\lambda), \qquad \mathcal{F}(x) = {}_2F_1(1/2, 1/2; 1; \lambda)^2 = \Theta. \tag{4}$$

The second identity is the classical quadratic transformation followed by Clausen's identity, with the branch specified by the expansion at $q = 0$. We use the determinant-normalized slash action $(f|_2 B)(\tau) = \det(B)(c\tau + d)^{-2} f(B\tau)$ for $B = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$.

2.1 The extra normalizer and its Hauptmodul

Let

$$\Gamma_0 = \Gamma_0(16), \qquad R = \begin{pmatrix} 1 & 1/2 \\ 8 & 5 \end{pmatrix}, \qquad W = \begin{pmatrix} 0 & -1 \\ 16 & 0 \end{pmatrix}, \qquad G = \langle \Gamma_0, R, W \rangle.$$

Matrices are interpreted projectively when acting on the upper half plane. This is not merely the usual Atkin–Lehner extension $\Gamma_0(16) + 16$: the additional coset represented by R is essential. In particular one cannot apply an Atkin–Lehner-only theorem without checking the argument for this extra normalizer.

Lemma 2.1. *Both R and W normalize Γ_0. The quotient G/Γ_0 has order four. Moreover*

$$x \circ R = x, \quad x \circ W = x, \qquad \Theta|_2 R = \Theta, \quad \Theta|_2 W = -\Theta.$$

The function x is a Hauptmodul for G. The zeros of Θ are precisely the two Γ_0-cusps with denominator 4, and these form one G-orbit. At either of these cusps the orders of x and Θ on $X_0(16)$ are -2 and 2, respectively.

Proof. The eta-quotient criterion applies at level 16. For x, the two exponent sums are 24 and 24, and its character is trivial. For Θ, the two sums are zero and its character is also trivial. Ligozat's cusp-order formula [9, Proposition 3.2.8] gives the following complete table. A denominator-c cusp has width $16/\gcd(c^2, 16)$.

c	1	2	4	8	16
number of cusps	1	1	2	1	1
$\mathrm{ord}_c(x)$	1	1	-2	1	1
$\mathrm{ord}_c(\Theta)$	0	0	2	0	0
width	16	4	1	1	1

The eta products have neither zeros nor poles in the upper half plane.

For an integral matrix $B = \left(\begin{smallmatrix} a & b \\ 16c & d \end{smallmatrix}\right)$ with odd determinant, direct multiplication gives

$$R^{-1}BR = \begin{pmatrix} 5a + 40b - 8c - 4d & (5a - 5d)/2 + 25b - 4c \\ 8(d - a) - 64b + 16c & -4a - 40b + 8c + 5d \end{pmatrix}. \tag{5}$$

Here a, d are odd, so all entries are integral and the lower left entry is divisible by 16. This proves the required inclusion for Γ_0. The reverse inclusion follows in the same way, or from

$$R^2 = \begin{pmatrix} 5 & 3 \\ 48 & 29 \end{pmatrix} \in \Gamma_0.$$

The normalization by W is the usual Fricke normalization. In addition $W^{-1}RW = R^{-1}$. Hence G/Γ_0 has at most four elements. The cosets represented by $1, R, W, RW$ are distinct: after scaling to determinant one, the latter three cannot be integral matrices in Γ_0. Thus the index is exactly four.

Conjugation by $D_4 = \mathrm{diag}(4, 1)$ sends Γ_0 to $\{\pm\gamma : \gamma \in \Gamma(4)\}$, sends W projectively to $S = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, and sends R to

$$R' = \begin{pmatrix} 1 & 2 \\ 2 & 5 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}.$$

For the theta expression in (3), each of the two factors on the right changes the sign of x, while their product preserves it. The transformation S interchanges ϑ_2 and ϑ_4, with the common automorphy factors canceling in x. The form $\vartheta_3(z)^4$ has trivial weight-two character on $\Gamma(2)$, whereas $\vartheta_3(-1/z)^4 = -z^2\vartheta_3(z)^4$. These facts prove the four transformation assertions.

The divisor table shows that $x : X_0(16) \to \mathbb{P}^1$ has degree four. It factors through the degree-four quotient by G/Γ_0, so the induced map $X(G) \to \mathbb{P}^1$ has degree one. This proves the Hauptmodul assertion. Finally $W(1/4) = -1/4$, so W exchanges the two denominator-4 cusp classes. □

3 The integral Hecke trace at every raw cutoff

For an odd prime p, put

$$M(16, p) = \left\{ \begin{pmatrix} a & b \\ 16c & d \end{pmatrix} : a, b, c, d \in \mathbb{Z},\ ad - 16bc = p \right\}.$$

Formula (5) proves $R^{-1}M(16,p)R = M(16,p)$; the same identity for W follows from its Fricke normalization. Thus conjugation by every element of G preserves this Hecke set and permutes its left Γ_0-cosets. These cosets have the usual representatives

$$\beta_\infty = \begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}, \qquad \beta_j = \begin{pmatrix} 1 & j \\ 0 & p \end{pmatrix} \quad (0 \le j < p).$$

For every $\beta \in M(16,p)$, the cusp $\beta(1/4)$ has denominator with 2-adic valuation exactly two after reduction to lowest terms: its numerator is odd and its denominator is 4 times an odd integer, and any cancellation factor divides the odd integer p. Hence it belongs to one of the two denominator-4 classes. Both have width one and belong to the same G-orbit. This is the cusp condition needed in the trace argument.

Let $P \in \mathbb{Z}[T]$. Define

$$Q_P(\tau) = \frac{\Theta(\tau)P(x(p\tau))}{\Theta(p\tau)}, \qquad G_P(q) = \frac{\Theta(q^p)P(x(q))}{\Theta(q)},$$

and, using the determinant-normalized slash operator,

$$R_\beta(\tau) = p\,\frac{\Theta(\tau)}{(\Theta|_2\beta)(\tau)}P(x(\beta\tau)).$$

The sum $\mathrm{Tr}_p(P) = \sum_{[\beta]} R_\beta$ is G-invariant. Indeed left multiplication by Γ_0 does not change a summand; right multiplication by Γ_0 permutes the summands; and for either normalizer $g = R, W$, the factors $\Theta|_2 g = \pm\Theta$ cancel to give $R_\beta(g\tau) = R_{g^{-1}\beta g}(\tau)$. Moreover

$$\mathrm{Tr}_p(P) = Q_P + p^3 U_p G_P, \qquad U_p\left(\sum b_n q^n\right) = \sum b_{pn} q^n. \tag{6}$$

For the p finite representatives, the factor is p^2 before summing, and the root-of-unity sum contributes the remaining factor p.

There are no poles in the upper half plane or at cusps outside the pole orbit of x. The latter assertion follows either directly from the denominator classes, or from the adjugate argument: if $\beta c'$ were in the pole orbit, then $p\beta^{-1}$ and normalizer conjugation would put c' in that orbit as well. At $c = 1/4$, precisely one Hecke branch multiplies the local order by p, and the other p branches divide it by p. This follows from Hermite normal form after a matrix sending ∞ to c. Equality of the cusp widths makes these the actual local scaling factors. Since the orders of Θ and x are 2 and -2, the unique possible largest pole has order at most $2(p-1) + 2p\deg P$. Consequently

$$\mathrm{Tr}_p(P) = \Pi_P(x), \qquad \Pi_P \in \mathbb{Z}[T], \qquad \deg \Pi_P \le p\deg P + p - 1. \tag{7}$$

Polynomiality follows from the Hauptmodul and the single pole orbit. Integrality follows by comparing the integral q-expansion in (6), using $x = q + O(q^2)$.

Let $q(X) \in X + X^2\mathbb{Z}[[X]]$ be the compositional inverse of $x(q)$, and define

$$\sigma_p(X) = x(q(X)^p), \qquad \rho_p(X) = \mathcal{F}(X)/\mathcal{F}(\sigma_p(X)).$$

The first series is congruent to X^p modulo p.

Theorem 3.1. *For every odd prime p and all $m, s \ge 1$,*

$$\rho_p(X)\mathcal{F}_{mp^{s-1}}(\sigma_p(X)) - \mathcal{F}_{mp^s}(X) \in p^{3s}\mathbb{Z}_p[[X]].$$

Proof. Fix m. Set $K_j = mp^j$, and interpret $\mathcal{F}_{K_{-1}}$ as zero. Apply (6)–(7) to $P = \mathcal{F}_{K_{s-1}}$. Its trace polynomial has degree strictly less than K_s, because $p(K_{s-1} - 1) + p - 1 = K_s - 1$. Assuming the assertion at level $s-1$, its q-form gives

$$G_P(q) \equiv \mathcal{F}_{K_{s-2}}(x(q^p)) \pmod{p^{3(s-1)}}.$$

Applying p^3U_p gives $p^3\mathcal{F}_{K_{s-2}}(x(q))$ modulo p^{3s}. For $s = 1$ the same conclusion holds because both sides are zero modulo p^3. Thus Q_P is congruent modulo p^{3s} to an integral polynomial in x of degree less than K_s. On the other hand

$$Q_P = \mathcal{F}(X)\frac{\mathcal{F}_{K_{s-1}}(\sigma_p(X))}{\mathcal{F}(\sigma_p(X))} \in \mathcal{F}(X) + X^{K_s}\mathbb{Z}[[X]].$$

Comparing coefficients below K_s identifies the polynomial modulo p^{3s} with $\mathcal{F}_{K_s}$. Coefficients at or above K_s vanish modulo p^{3s}, as required. This also starts the induction. □

4 Integral specialization on the ordinary modular curve

We now restrict to $p \equiv 1 \pmod 4$. The use of the ordinary modular curve in this section is essential: the second Hasse–Witt factor of the cubic parameter vanishes at $64X^2 = 1$, so the original BV3 ring that inverts that factor cannot be evaluated there.

Conjugating by D_4 identifies $X_0(16)$ analytically with the full-level curve $X(4)$ (with ± 1 identified). Since $p \equiv 1 \pmod 4$, choose a fourth root of unity in $\mathbb{Z}_p$; the corresponding component of $X(4)$ has good reduction and a Tate cusp with local parameter $q = e^{2\pi iz/4}$. The functions and forms in (3) are defined on this component. They have coefficients in $\mathbb{Z}_p$.

For a precise integral chart, let V be the smooth compactified level-four component over $\mathbb{Z}_p$ with the two pole-cusp sections removed, and let $A_0 = \mathcal{O}(V)$. This is affine. The functions have no vertical poles at odd primes: for x use (4) and normality; for Θ use

$$\Theta^6 = \frac{\Delta(4\tau)^5}{\Delta(2\tau)^2\Delta(8\tau)^2}.$$

The curves in this identity are related by prime-to-p isogenies, and their discriminants do not vanish on smooth elliptic fibers. At retained cusps the leading constants are powers of two and roots of unity. Consequently Θ trivializes the weight-two Hodge line bundle on V in characteristic zero and in characteristic p.

Let Ha denote the mod-p Hasse invariant of weight $p-1$. The quotient $\mathrm{Ha}\,/\Theta^{(p-1)/2}$ is a function on $V_{\mathbb{F}_p}$. Choose a lift $h_4 \in A_0$ and put

$$A = A_0[1/h_4], \qquad \widehat{A} = \varprojlim_n A/p^nA.$$

Its special fiber is the ordinary locus of V and is geometrically integral. At the Tate cusp h_4 is a unit, so q-expansions lie in $\mathbb{Z}_p[[q]]$. The resulting formal open is stable under the canonical quotient, since its reduction is Frobenius. A CM level marking may require a finite unramified extension; all the following arguments remain valid after that base change.

The following standard facts about the ordinary modular curve are used in their integral form.

1. Quotienting by the canonical connected subgroup of order p gives an integral Frobenius lift Φ on the ordinary formal curve. On the Tate parameter it sends q to q^p. It carries prime-to-p level structures along the quotient and reduces to Frobenius modulo p.

2. If $\phi : E \to E'$ is the canonical isogeny, then $p^{-1}\phi^* : \omega_{E'} \to \omega_E$ is an isomorphism on the ordinary formal locus. For the weight-two form Θ, its ratio with the image of $\Phi^*\Theta$ under this normalized map is an integral unit $\rho \in \widehat{A}^\times$, with q-expansion $\Theta(q)/\Theta(q^p)$.

3. The q-expansion homomorphism is injective modulo p on A/pA. Indeed a rational function on the integral special-fiber modular curve whose expansion at a smooth cusp is zero is identically zero. Consequently, for every $e \geq 0$,

$$\widehat{A} \cap p^e\mathbb{Z}_p[[q]] = p^e\widehat{A}. \tag{8}$$

The higher powers follow by successive division by p. This is the integral q-expansion principle, not an assertion about convergence of the individual hypergeometric period at a unit argument.

These are the standard canonical-subgroup and q-expansion constructions on the ordinary modular curve: see Katz, Proposition 2.7.1 for divisibility detected by a q-expansion, Theorems 3.1 and 3.3 and Section 3.4 for the ordinary canonical quotient, and Appendix A2.1–A2.2 for the normalized differential and its Tate-curve expansion [5]. They can also be made on a finer auxiliary prime-to-p level and then descended.

The pole cusps and the ordinary locus are preserved under reduction of Φ, so Φ^*x is integral on this formal open. Its q-expansion is $x(q^p)$; therefore it realizes the same formal σ_p as above. All terms of Theorem 3.1 belong to $\widehat{A}$. Applying (8) to their q-expansions yields

$$\rho\,\mathcal{F}_{mp^{s-1}}(\Phi^*x) - \mathcal{F}_{mp^s}(x) \in p^{3s}\widehat{A}. \tag{9}$$

Thus the full exponent $3s$ is retained under evaluation at every point of this ordinary formal open. No assertion about convergence of $\mathcal{F}(1/8)$ is required.

4.1 The ordinary CM point and its canonical branch

At $z = i$, equivalently $\tau = i/4$, one has $\lambda(i) = 1/2$ and $x(i/4) = 1/8$. The sign follows from the positive real theta constants. The elliptic curve in the z coordinate has complex multiplication by $\mathbb{Z}[i]$; at $p \equiv 1 \pmod 4$ it has good ordinary reduction. A full level-four marking can be chosen over a finite unramified extension with integer ring $\mathcal{O}$.

Write $p = a^2 + b^2$, with a odd and b even, and choose the Gaussian generator β of the prime above p which has positive valuation in the chosen embedding. Its kernel has connected reduction, so the isogeny given by β is the canonical connected quotient at this CM lift. Since $\beta \equiv 1 \pmod 2$, it fixes the full level-two structure. After the target is identified with the same CM elliptic curve, its Legendre parameter is therefore still $\lambda = 1/2$.

Applying Φ^* to $16x^2 = \lambda(1-\lambda)$ now gives

$$(\Phi^*x)^2 = 1/64$$

at the CM point. The integral Frobenius lift reduces to $x \mapsto x^p$, so $\Phi^*x \equiv 1/8 \pmod p$. Since p is odd, the two roots $1/8$ and $-1/8$ are distinct modulo p. Consequently

$$(\Phi^*x)(i/4) = 1/8. \tag{10}$$

This proves fixedness on the required canonical branch.

For an explicit check on the finer level structure, let $\alpha = a + bi$ be the unit generator and $\beta = a - bi$, so that $\alpha\beta = p$. The matrix of multiplication by β on level-four homology, up to transpose and sign convention, reduces modulo 4 to $\pm I$ or

$$\pm R' \equiv \pm \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}.$$

Indeed a is odd, and b is either 0 or 2 modulo 4. Both matrices preserve x and have eigenvalue $+1$ on Θ. If ω is the differential after identifying the target with the same CM curve, the normalized target differential is $\omega' = (p/\beta)\omega = \alpha\omega$, since $\phi^*\omega' = p\omega$. Weight two means that $\Theta(E, \alpha\omega) = \alpha^{-2}\Theta(E, \omega)$. Hence

$$\rho(i/4) = \alpha^2. \tag{11}$$

The main conclusion below requires only that this value is a single fixed unit L_p; its exact gamma normalization will follow from Guo's published theorem.

All CM statements here concern the maximal-order curve at $z = i$. The original lattice with period ratio $\tau = i/4$ has conductor four, and an unrestricted norm-p endomorphism assertion on that lattice would be incorrect. The level-four model and the canonical quotient give the fixedness statement actually used.

5 The ordinary gamma multiplier and arbitrary-length recurrence

Let L_p be the value of ρ at the ordinary CM point. The previous section proves that L_p is a fixed p-adic unit, independent of m, s, and that for every $m, s \geq 1$

$$A_{m,s} \equiv L_p A_{m,s-1} \pmod{p^{3s}}. \tag{12}$$

One may compute L_p as the Gaussian unit-root square in (11). For the gamma normalization, there is a shorter route which uses only the published weaker recurrence.

Lemma 5.1. *For $p \equiv 1 \pmod 4$, each H_s is a p-adic unit, and $H_s \equiv H_1^s \pmod p$.*

Proof. Put $d = (p-1)/2$. The coefficient $c_k = \binom{2k}{k}^3/64^k$ is zero modulo p if a base-p digit of k exceeds d. Otherwise Lucas's congruence and Fermat's theorem give $c_k \equiv \prod_j c_{k_j} \pmod p$, where k_j are those digits. Every digit string with $0 \leq k_j \leq d$ represents an integer at most $(p^s-1)/2$. Hence the half-length sum factors as H_1^s modulo p. Guo's published first-level case gives $H_1 \equiv -\Gamma_p(1/4)^4 \pmod p$, which is a unit. □

Lemma 5.2. *The constant L_p in (12) equals $-\Gamma_p(1/4)^4$ exactly in $\mathbb{Z}_p$.*

Proof. Write $g_p = -\Gamma_p(1/4)^4$. Guo's theorem [2, arXiv equation (1.9)] states

$$H_s \equiv g_p H_{s-1} \pmod{p^{s+1}} \qquad (s \geq 1).$$

The case $m = 1$ of (12) gives the same recurrence with L_p and modulus p^{3s}. Subtracting and using Lemma 5.1 therefore gives $L_p - g_p \in p^{s+1}\mathbb{Z}_p$ for every s. The intersection of these ideals is zero. This proves the exact equality. If the CM marking was evaluated over an unramified extension with integer ring $\mathcal{O}$, run this argument first in $\mathcal{O}$. It proves the same equality and hence descent to $\mathbb{Z}_p$. □

Proof of the ordinary assertion in Theorem 1.1. Substitute Lemma 5.2 into (12). This gives the assertion with modulus p^{3s}. Write $m = p^v m_0$, with $p \nmid m_0$. Then $A_{m,s} = A_{m_0,s+v}$ and $A_{m,s-1} = A_{m_0,s+v-1}$. Applying the same assertion at level $s+v$ gives the displayed stronger modulus $p^{3s+3v_p(m)}$. For $m = 1$ the upper summation index is exactly $(p^s-1)/2$; for $m = 2$ it is $p^s - 1$. Thus both stated truncation conventions follow. □

6 Integral Legendre jets and change of Frobenius lift

Throughout this section, p may be any odd prime. The local ring notation is reset on the punctured Legendre base; the restriction $p \equiv 1 \pmod 4$ in Section 4 is not imposed here. Write

$$D = \lambda(1-\lambda), \qquad h = 1 - 2\lambda, \qquad A = \widehat{\mathbb{Z}_p[\lambda, D^{-1}]},$$

where completion is p-adic. Put

$$N_s = (p^s-1)/2, \qquad P_s(\lambda) = \sum_{k=0}^{N_s} \binom{2k}{k}^3 (D/16)^k, \qquad Y_s = (P_s, P_s', P_s'')^{\mathsf{T}}.$$

The desired half-length sum is

$$H_s = P_s(1/2) = \sum_{k=0}^{(p^s-1)/2} \frac{(1/2)_k^3}{k!^3}.$$

The quadratic pullback is essential: the differential equation in the raw cubic parameter is singular at $X = 1/8$, whereas the connection used below is regular at $\lambda = 1/2$.

The formal input is Theorem 3.1, proved above by the raw Hecke trace. In the present notation, on the ordinary locus it gives

$$P_s(\lambda) - \rho(\lambda) P_{s-1}(\sigma(\lambda)) \in p^{3s} A^{\mathrm{ord}}, \tag{13}$$

where σ is the ordinary canonical quotient lift and $\rho = (L/L^\sigma)^2$, with $L = {}_2F_1(1/2, 1/2; 1; \lambda)$. This is an identity of integral analytic quotients; the individual series L need not converge at unit arguments. Here

$$A^{\mathrm{ord}} = \mathbb{Z}_p[\lambda, \widehat{D^{-1}}, \widehat{\mathscr{H}^{-1}}]$$

and $\mathscr{H}$ is any integral lift of the nonzero Legendre Hasse polynomial; both hats denote p-adic completion. Equivalently it is the p-adic completion of the localization of A at $\mathscr{H}$. To obtain (13), substitute $X^2 = D/16$ in the even raw truncations and use the ordinary q-expansion principle. The corresponding ordinary Frobenius lifts commute with this modular pullback. All terms are integral on the ordinary locus, and divisibility of their cusp expansions gives the stated divisibility in A^{ord}; the integral ordinary q-expansion principle is Katz's Proposition 2.7.1 [5], and the canonical quotient is constructed in his Theorems 3.1 and 3.3 and Section 3.4. The raw Tate parameter satisfies $q_E = q_{\mathrm{raw}}^4$, so its replacement by q_{raw}^p is exactly $q_E \mapsto q_E^p$ on the Legendre Tate curve. The fact that the raw truncations are even makes the substitution $X^2 = D/16$ integral without choosing a square root of λ.

6.1 An exact differential-equation residual

Set

$$P = \frac{h}{D}, \qquad Q = -\frac{1}{4D}.$$

The Legendre equation is $y'' + Py' + Qy = 0$. Its symmetric-square equation has monic operator

$$\mathcal{L} = \partial^3 + 3P\partial^2 + (P' + 4Q + 2P^2)\partial + (2Q' + 4PQ).$$

Equivalently,

$$D^2\mathcal{L} = D^2\partial^3 + 3Dh\partial^2 + (1 - 7D)\partial - h/2. \tag{14}$$

Lemma 6.1. *For every $s \geq 0$,*

$$\mathcal{L}P_s = -\frac{p^{3s}}{2} \frac{\binom{2N_s}{N_s}^3}{16^{N_s}} h D^{N_s - 2} \in p^{3s} A.$$

In particular the jet vector satisfies $Y_s' = CY_s + r_s e_3$, where C is the integral companion matrix of $\mathcal{L}$, $r_s \in p^{3s}A$, and $e_3 = (0, 0, 1)^{\mathsf{T}}$.

Proof. Direct differentiation gives

$$(D^2\mathcal{L})(D^k) = h\left(k^3 D^{k-1} - \frac{(2k+1)^3}{2} D^k\right).$$

For $a_k = \binom{2k}{k}^3/16^k$ one has $k^3 a_k = (2k-1)^3 a_{k-1}/2$. All interior terms therefore cancel, and the last term is $-(2N_s + 1)^3 a_{N_s} h D^{N_s}/2$. Since $2N_s + 1 = p^s$, division by the unit D^2 in A proves the claim. □

6.2 The integral rank-three Frobenius matrix

We use the constant quadratic twist

$$E_\lambda^- : \quad y^2 = -x(x-1)(x-\lambda).$$

At $\lambda = 0$ its nodal tangents are ± 1, so its Tate degeneration is split. This makes the following weight-one normalization unambiguous. For the usual Legendre curve the weight-one matrix differs by the constant sign $(-1/p)$; its symmetric square is identical. This sign and the usual Legendre Frobenius normalization are recorded explicitly in Shirai's introduction, equations (1), (3), and (4) [7].

Let $\omega = dx/y$ and $\eta = \nabla_{\partial_\lambda}\omega$. These form an integral basis of $H^1_{\mathrm{dR}}(E_\lambda^-/A)$. Indeed in the usual basis $(\omega, x\omega)$ one has

$$\eta = \frac{\omega}{2(1-\lambda)} - \frac{x\omega}{2\lambda(1-\lambda)}, \tag{15}$$

whose determinant is a unit for every odd p. The relation follows by reducing the exact differential $d(x(x-1)/y)$. The Gauss–Manin connection is

$$\nabla\begin{pmatrix}\omega\\ \eta\end{pmatrix} = \begin{pmatrix}0 & 1\\ -Q & -P\end{pmatrix}\begin{pmatrix}\omega\\ \eta\end{pmatrix} d\lambda.$$

Let $\phi : A \to A$ be any lift of Frobenius. The integral crystalline Frobenius of this smooth proper elliptic family gives a matrix M_ϕ with entries in A. Our convention is a row convention:

$$F_\phi(\phi^*(\omega,\eta)^{\mathsf{T}}) = M_\phi(\omega,\eta)^{\mathsf{T}}.$$

Its determinant is p times a unit, by the cup product and the action of Frobenius on H^2. Consequently

$$V_\phi = pM_\phi^{-1} \in M_2(A). \tag{16}$$

This uses the adjugate of a rank-two matrix and does not invert the Hasse invariant. For the underlying integral comparison, one can use the Dieudonne crystal of the elliptic scheme's p-divisible group: Berthelot–Messing [6, §2, Theorem 2, p. 23] gives its local freeness, the construction on pp. 24–25 gives integral F, V with $FV = VF = p$, and [6, §3, equation (3.1), p. 27] identifies it with relative H^1_{cris}, compatibly with the Hodge filtration. Its evaluation on the smooth lift is integral de Rham cohomology. For the completed comparison and the canonical integral Frobenius map, see also [8, Remark 60.24.14, Tag 07MZ; Theorem 60.26.4, Tag 07N5]; the latter map is constructed before inverting p. The first Hodge line is carried into pH^1 by Frobenius: modulo p this follows from the pullback of regular differentials under relative Frobenius being zero, together with the integral comparison and its functoriality. In the row convention this says that the first row of M_ϕ is divisible by p at a Frobenius-fixed fiber.

The symmetric-square jet basis is

$$\omega^2, \qquad 2\omega\eta, \qquad 2\eta^2 - 2P\omega\eta - 2Q\omega^2.$$

Its change-of-basis matrix from $(\omega^2, \omega\eta, \eta^2)$ is

$$S = \begin{pmatrix}1 & 0 & 0\\ 0 & 2 & 0\\ -2Q & -2P & 2\end{pmatrix}, \qquad \det S = 4. \tag{17}$$

Define the integral period-side Frobenius matrix

$$B_\phi = S\,\mathrm{Sym}^2(V_\phi)(S^\phi)^{-1} \in M_3(A). \tag{18}$$

The word "period-side" records the inverse in (16); confusing this matrix with the cohomology-row matrix would reverse the required Hodge divisibility.

On the ordinary canonical lift σ, write $a = L/L^\sigma$. Here the exact differential normalization deserves to be specified. The connected subgroup quotient reduces to relative Frobenius; transporting the ordered 2-torsion gives the parameter $\sigma(\lambda)$. Its model-scaling unit relative to $E^-_{\sigma(\lambda)}$ is congruent to one modulo p and therefore has a unique square root congruent to one for odd p. This identifies the quotient pullback with the crystalline F_σ, without an unspecified quadratic twist. One may make this identification on a finite etale auxiliary level and descend; the matrices in our displayed basis already lie over A.

On the split Tate chart, the differential satisfies $\omega = L(\lambda)\omega_{\mathrm{Tate}}$, with constant term one. This identity follows either from the normalized Tate period or from its Legendre differential equation and its initial value at $\lambda = 0$. The canonical isogeny sends $z \mapsto z^p$ and hence pulls back ω_{Tate} to $p\omega_{\mathrm{Tate}}$ [5, Appendix A2, especially (A2.2.1) and (A2.2.7), pp. 175–177]. Consequently its differential is exactly

$$\omega^\sigma \longmapsto c\omega, \qquad c = p\frac{L^\sigma}{L} = \frac{p}{a}.$$

The identity on the whole ordinary ring follows from the integral q-expansion principle; thus no parameter-dependent unit is being suppressed. Horizontality now gives

$$M_\sigma = \begin{pmatrix} c & 0 \\ c'/\sigma' & c/\sigma' \end{pmatrix}, \qquad pM_\sigma^{-1} = \begin{pmatrix} a & 0 \\ a' & a\sigma' \end{pmatrix}.$$

These equalities can be read in the fraction field; the resulting matrices are the integral ones above. Taking the symmetric square in the jet basis yields

$$B_\sigma = \begin{pmatrix} \rho & 0 & 0 \\ \rho' & \rho\sigma' & 0 \\ \rho'' & 2\rho'\sigma' + \rho\sigma'' & \rho(\sigma')^2 \end{pmatrix}, \qquad \rho = a^2. \tag{19}$$

In particular its columns have respective valuations at least $0, 1, 2$, because $\sigma' \equiv \sigma'' \equiv 0 \pmod p$. Differentiating (13) twice therefore gives

$$Y_s - B_\sigma Y_{s-1}^\sigma \in (p^{3s}A^{\mathrm{ord}})^3. \tag{20}$$

6.3 Changing the lift without losing three digits

Recall the definition of ϵ_p from Section 1.

Lemma 6.2 (Taylor transport with a third-order residual)**.** *Let R be one of the above p-adically complete differential rings, with $\partial R \subseteq R$, and let $u, v : R \to R$ be substitutions congruent modulo p. Suppose $Y = (f, f', f'')^{\mathsf{T}}$ and $Y' = CY + re_3$, with $r \in p^bR$ and the integral companion matrix C. If $T(u, v)$ is horizontal Taylor transport from v to u, then*

$$Y^u - T(u, v)Y^v \in p^{b+3-\epsilon_p}R \ \oplus \ p^{b+2}R \ \oplus \ p^{b+1}R.$$

Proof. Put $\delta = u(\lambda) - v(\lambda) \in pR$. The matrices of iterated covariant differentiation have integral entries. Differentiating the inhomogeneous companion system repeatedly shows that its residual first enters the three jet coordinates at derivative orders $3, 2, 1$, respectively. Subsequent coefficients are integral combinations of the coefficients of C, their derivatives, and derivatives of r; hence they remain in p^bR.

Subtract the horizontal Taylor series from the ordinary Taylor series for Y. The resulting three sums start at orders $3, 2, 1$ and their nth terms are divisible by $p^b\delta^n/n!$. They converge since $v_p(\delta^n/n!) \geq n - v_p(n!) \to \infty$ for odd p. For the three starting orders, the respective lower bounds are

$$\min_{n\geq 3}(n - v_p(n!)) = 3 - \epsilon_p, \quad \min_{n\geq 2}(n - v_p(n!)) = 2, \quad \min_{n\geq 1}(n - v_p(n!)) = 1.$$

This proves the assertion and identifies the unique extra loss at $p = 3$. □

Proposition 6.3. *For any integral Frobenius lift ϕ of the punctured Legendre base and all $s \geq 1$,*

$$Y_s - B_\phi Y_{s-1}^\phi \in (p^{3s-\epsilon_p} A)^3.$$

Proof. First work on the ordinary localization. The crystalline change-of-lift isomorphism is horizontal Taylor transport; see [6, §2.1, pp. 19–20] and [8, Proposition 60.17.4, Tag 07JH], where the divided-power Taylor formula and its cocycle identity are explicit. In the rank-two row convention, if $T_2(\sigma, \phi)$ transports solution coordinates from ϕ to σ, the cohomology matrices satisfy

$$M_\phi = T_2(\sigma, \phi)^{-1} M_\sigma, \qquad V_\phi = V_\sigma T_2(\sigma, \phi).$$

Taking symmetric squares and changing to the jet bases gives precisely $B_\phi = B_\sigma T(\sigma, \phi)$. Apply Lemma 6.2 to Y_{s-1}, using $b = 3(s-1)$ from Lemma 6.1. Multiplication by B_σ adds at least $0, 1, 2$ to its three columns. Therefore

$$B_\sigma (Y_{s-1}^\sigma - T(\sigma, \phi) Y_{s-1}^\phi) \in (p^{3s-\epsilon_p} A^{\mathrm{ord}})^3.$$

Combine this with (20).

Both Y_s and $B_\phi Y_{s-1}^\phi$ already belong to A^3, since (18) is integral on the whole smooth Legendre base. The map $A/pA \to A^{\mathrm{ord}}/pA^{\mathrm{ord}}$ is injective: the latter is localization at the nonzero Hasse polynomial in the integral domain $\mathbb{F}_p[\lambda, D^{-1}]$. Successive division by p then gives $A \cap p^e A^{\mathrm{ord}} = p^e A$ for every e. This descends the congruence without any additional loss. □

7 Supersingular CM specialization

Assume now $p \equiv 3 \pmod 4$ and choose

$$\phi(\lambda) = \frac{1}{2} + \left(\lambda - \frac{1}{2}\right)^p.$$

It is an integral Frobenius lift, preserves invertibility of D modulo p, and fixes $\lambda_0 = 1/2$. The elliptic curve $E_{1/2}^-$ has good supersingular reduction and Frobenius trace zero. For example, it is a quadratic twist of $y^2 = x^3 - x$; at $p \equiv 3 \pmod 4$ the trace of that curve is zero by pairing x with $-x$ in its quadratic-character sum. Let $M = M_\phi(1/2)$. Its characteristic polynomial is $T^2 + p$, so

$$M^2 = -pI_2, \qquad V = pM^{-1} = -M.$$

Frobenius–Hodge divisibility says that the first row of M is in $p\mathbb{Z}_p^2$. Consequently, for $B = B_\phi(1/2)$,

$$B^2 = p^2 I_3, \qquad (1, 0, 0)B \in p^2 \mathbb{Z}_p^3. \tag{21}$$

The first assertion follows by taking a symmetric square. For the second, the first symmetric-square basis vector is ω^2, and the first row of S in (17) is $(1, 0, 0)$; multiplying by the integral inverse of S preserves divisibility by p^2.

Theorem 7.1. *For every prime $p \equiv 3 \pmod 4$ and every integer $s \geq 2$,*

$$H_s - p^2 H_{s-2} \in p^{3s-1-\epsilon_p} \mathbb{Z}_p.$$

Thus the exponent is $3s - 1$ for $p \geq 7$, and $3s - 2$ for $p = 3$.

Proof. Evaluate Proposition 6.3 at λ_0 and write

$$Y_s(\lambda_0) = B Y_{s-1}(\lambda_0) + E_s, \qquad E_s \in (p^{3s-\epsilon_p} \mathbb{Z}_p)^3.$$

Applying this at two consecutive levels gives

$$Y_s(\lambda_0) - p^2 Y_{s-2}(\lambda_0) = E_s + B E_{s-1}.$$

By (21), the first coordinate of the second term is divisible by $p^{2+3(s-1)-\epsilon_p} = p^{3s-1-\epsilon_p}$; the first term has one more factor of p. The first coordinate of $Y_s(\lambda_0)$ is H_s, proving the asserted two-step recurrence. □

7.1 A coupled auxiliary-sum formulation

For $s \geq 0$ put

$$W_s = \sum_{k=0}^{(p^s-1)/2} (4k+1)\frac{(1/2)_k^3}{k!^3} = H_s - \frac{1}{2}P_s''(1/2).$$

The second equality follows by differentiating $D = \lambda(1-\lambda)$ twice at $\lambda = 1/2$: $D' = 0$, $D'' = -2$, and $D = 1/4$.

Corollary 7.2. *For each $p \equiv 3 \pmod 4$ there is a unit $\mu_p \in \mathbb{Z}_p^\times$, the square of an integral crystalline matrix coefficient, such that for every $s \geq 1$,*

$$H_s \equiv -\frac{p^2}{\mu_p} W_{s-1} \pmod{p^{3s-\epsilon_p}},$$
$$W_s \equiv -\mu_p H_{s-1} \pmod{p^{3s-\epsilon_p}}.$$

No special-function identification of μ_p is needed or asserted.

Proof. At $\lambda = 1/2$, equation (15) gives $\eta = (1-2x)\omega$. After adjoining i, the CM automorphism $(x, y) \mapsto (1-x, iy)$ acts on (ω, η) as $\mathrm{diag}(i, -i)$. Since $p \equiv 3 \pmod 4$, crystalline Frobenius conjugates i to $-i$. Its matrix, and hence $V = -M$, is off diagonal:

$$V = \begin{pmatrix} 0 & b \\ c & 0 \end{pmatrix}, \qquad bc = -p.$$

The Hodge-row divisibility and integrality give $v_p(b) = 1$ and $v_p(c) = 0$. Set $\mu_p = c^2$; then $b^2 = p^2/\mu_p$.

At this point $P = 0$ and $Q = -1$, so in symmetric-square solution coordinates the first and third entries of the vector corresponding to Y_s are

$$H_s, \qquad \tfrac{1}{2}P_s''(1/2) - H_s = -W_s.$$

The middle entry is zero because $P_s(\lambda) = P_s(1-\lambda)$. The symmetric square of the displayed off-diagonal matrix interchanges the first and third coordinates with factors b^2 and c^2. Proposition 6.3, with the integral basis change (17), therefore gives the two congruences with the same modulus. □

In particular these congruences prove the exact valuations

$$v_p(H_s) = 2\lceil s/2 \rceil, \qquad v_p(W_s) = 2\lfloor s/2 \rfloor.$$

For $p \geq 7$ this follows immediately by induction from $H_0 = W_0 = 1$, because each error term has strictly higher valuation than the displayed leading term. For $p = 3$ the same induction starts from the explicit values $H_1 = 9/8$ and $W_1 = 13/8$; at all later levels the error valuation is again strictly higher. In particular $H_1 \in p^2\mathbb{Z}_p$. Together with Theorem 7.1, this proves the supersingular assertion in Theorem 1.1, including its first level.

What supplies the exponent. The scalar ordinary theorem alone cannot be evaluated at this point. The proof uses its differentiated rank-three form, the exact p^{3s} differential-equation residual, and weighted Taylor transport to an integral crystalline matrix that is defined on the supersingular fiber. The loss of one digit comes from combining the previous-level error p^{3s-3} with the Hodge-row factor p^2. The additional loss at $p = 3$ comes from $3!$ in the order-three Taylor term. Frobenius slopes or the identity $B^2 = p^2 I$ by themselves do not supply either estimate.

8 Precision, scope, and exact computational checks

Theorem 1.1 addresses the half-length sums at every stated level for all odd primes, with exponent $3s-2$ at $p=3$. For ordinary primes it also gives arbitrary raw lengths, including the full cutoff. The supersingular proof uses $2N_s+1=p^s$ in its exact differential residual; an extension to another cutoff would require a separate residual estimate.

The two losses in the supersingular exponent have different sources. The one-step jet congruence has precision $3s-\epsilon_p$. In the two-step composition, the preceding error has precision $3(s-1)-\epsilon_p$, and the first-row Hodge factor restores two digits. This gives $3s-1-\epsilon_p$. The extra loss at $p=3$ occurs in $\delta^3/3!$ in Taylor transport. Both estimates are part of the proof, independently of the numerical patterns below.

8.1 Ordinary half- and full-cutoff checks

The ordinary verification program checks the multiplier as the Gaussian unit root of

$$X^2-2(a^2-b^2)X+p^2, \qquad p=a^2+b^2, \quad a \text{ odd}.$$

It checks 23 prime/level pairs, in both half and full cutoffs, at precision p^{3s+3}. Every tested half- and full-cutoff defect has valuation exactly $3s$.

Prime	Levels s	Observed half-sum defect valuations
5	$1,\dots,6$	$3,6,9,12,15,18$
$13,17$	$1,\dots,4$	$3,6,9,12$
$29,37,41$	$1,2,3$	$3,6,9$

At $p=5$, the upper-half tail at level two has valuation four, below the target six, whereas its recurrence defect has valuation seven in the exact calculation. The raw-cutoff argument handles both sums without requiring that tail itself to vanish at the target precision.

8.2 Supersingular half-cutoff checks

The supersingular program works at precision $p^{3s_{\max}+5}$ and removes p-power factors before every division. The table lists $v_p(H_s-p^2H_{s-2})$ in increasing order of s. Each displayed valuation is exact at the working precision.

Prime	Levels s	Observed two-step defect valuations
3	$2,\dots,10$	$4,7,10,13,16,19,22,25,28$
7	$2,\dots,6$	$5,8,11,14,17$
11	$2,\dots,5$	$5,8,11,14$
$19,23$	$2,3,4$	$5,8,11$
$31,43,47$	$2,3$	$5,8$

These are 30 prime/level recurrence checks. They establish sharpness in the tested cases, not equality of the defect valuation for every prime and level.

8.3 Formal checks and reproducibility

A third program checks the raw formal identity directly from the eta products. Its 13 parameter sets include primes $3,5,7,11$, several values of m, and levels through $s=4$. Computations use exact modular arithmetic, with an explicit bound preventing integer overflow in the NumPy convolutions. Each parameter set checks only the finite coefficient window specified in the program.

All three programs are reproduced in Appendix B. The two numerical recurrence programs use only Python's standard library; the eta-product program also requires NumPy. The combined

evidence consists of 23 ordinary prime/level pairs (each checked in two cutoffs), 30 supersingular recurrence pairs, and 13 finite formal parameter sets. These categories are distinct. The general claims are supplied by the preceding arguments, rather than by an extrapolation from the computations.

A An independent finite identity and valuation bound

Fix an odd prime $p \equiv 3 \pmod 4$. Write

$$b_j = \frac{(1/2)_j}{j!} = \frac{\binom{2j}{j}}{4^j}, \qquad q_s = p^s, \qquad n_s = (q_s - 1)/2, \qquad H_s = \sum_{k=0}^{n_s} b_k^3.$$

For an odd positive integer $t \equiv 1 \pmod 4$, put

$$C(t) = b_{(t-1)/4}^2, \qquad U_s = \frac{\binom{p^s-1}{(p^s-1)/2}}{4^{p^s-1}}.$$

This appendix supplies an independent finite identity and valuation bound. It is not needed for the supersingular argument in Sections 6–7.

A.1 Exact interpolation identity

Proposition A.1. *For every integer $n \geq 0$, with $q = 2n + 1$,*

$$\sum_{k=0}^{n} b_k^3 = q^2 b_n^2 \sum_{l=0}^{\lfloor n/2 \rfloor} \frac{b_l^2 (n!)^2}{(4l+1)(n-2l)!(n+2l+1)!} \tag{22}$$

$$= q \frac{\binom{2n}{n}}{16^n} \sum_{l=0}^{\lfloor n/2 \rfloor} \binom{q}{n-2l} \frac{b_l^2}{4l+1}. \tag{23}$$

Proof. Consider the polynomial in T^2 of degree n

$$\mathcal{A}_n(T) = \sum_{k=0}^{n} b_k^3 \prod_{j=0}^{k-1} \left(1 - \frac{T^2}{(2j+1)^2}\right).$$

Let P_j denote the classical Legendre polynomial of degree j. For $0 \leq j \leq n$, the terminating Clausen identity gives

$$\mathcal{A}_n(2j+1) = {}_3F_2\left(\begin{matrix} -j, j+1, 1/2 \\ 1, 1 \end{matrix}; 1\right) = \mathsf{P}_j(0)^2.$$

This value is zero for odd j, and is $b_{j/2}^2$ for even j. Lagrange interpolation at the nodes $(2j+1)^2$ gives the coefficient

$$L_j(0) = \prod_{\substack{0 \leq i \leq n \\ i \neq j}} \frac{-(2i+1)^2}{(2j+1)^2 - (2i+1)^2} = \frac{(-1)^j ((2n+1)!!)^2}{4^n (2j+1)(n-j)!(n+j+1)!}.$$

Only $j = 2l$ contributes. Since $((2n+1)!!)^2/4^n = q^2 b_n^2 (n!)^2$, this proves (22). Using $q^2 b_n^2 (n!)^2/q! = q\binom{2n}{n}/16^n$ proves (23). □

In particular, define the finite index set

$$\mathcal{T}_s = \{t \in \mathbb{Z} : 0 < t \leq p^s,\ t \equiv 1 \pmod 4\}.$$

Then the exact identity becomes

$$H_s = \sum_{t \in \mathcal{T}_s} \mathcal{K}_s(t), \qquad \mathcal{K}_s(t) = p^s U_s \binom{p^s}{(p^s - t)/2} \frac{C(t)}{t}. \tag{24}$$

Every quantity in this equation is rational; no convergence or analytic continuation is used.

A.2 Carries and unconditional divisibility

Lemma A.2. *If $t = 4l + 1 \in \mathcal{T}_s$ and $a = v_p(t)$, then*

$$v_p(\mathcal{K}_s(t)) = 2s + 2v_p\binom{2l}{l} - 2a, \qquad v_p\binom{2l}{l} \geq \lceil a/2 \rceil.$$

Consequently,

$$v_p(\mathcal{K}_s(t)) \geq 2s - 2\lfloor a/2 \rfloor, \qquad v_p(H_s) \geq 2\lceil s/2 \rceil.$$

For even s, equality holds: $v_p(H_s) = s$.

Proof. U_s is a p-adic unit. If $t < p^s$ and $k = (p^s - t)/2$, then $v_p(k) = a$ and $v_p\binom{p^s}{k} = s - a$. The latter follows from $\binom{p^s}{k} = (p^s/k)\binom{p^s-1}{k-1}$ and Lucas's theorem. For $t = p^s$, the same formula gives $s - a = 0$. Equation (24) gives the exact valuation asserted.

For the carry bound, write $p = 4h + 3$. The base-p digits forced by $4l + 1 \equiv 0 \pmod{p^a}$ alternate, from the least significant end, between $3h + 2$ and h. If a is odd, the last forced digit is $3h + 2$; if a is even, it is h. In adding $l + l$, each $3h + 2$ digit produces a carry, and each following h digit, with its incoming carry, produces none. Thus at least $\lceil a/2 \rceil$ carries occur. Kummer's theorem proves the bound.

For odd s, $p^s \equiv 3 \pmod 4$, so $p^s \notin \mathcal{T}_s$ and $a \leq s - 1$. This gives valuation at least $s + 1$. For even s, the only term with $a = s$ is $t = p^s$, and its carry count is exactly $s/2$. Its valuation is s, whereas every other term has valuation at least $s + 2$. There can be no cancellation of this unique least-valued term. □

B Reproducible verification programs

Each listing is a separate executable source file. Save it under the specified name and run it with Python 3; it writes a JSON record next to its source. Listings B.1 and B.3 use only the standard library. Listing B.2 additionally requires NumPy. Their finite scope is described in Section 8.

B.1 Ordinary half- and full-cutoff recurrences

Save as `check_recurrences.py`.

```
"""Finite exact checks; the all-level theorem is proved in the paper."""
from math import isqrt, comb
from pathlib import Path
import json

def strip(n, p):
    v = 0
    while n % p == 0:
        n //= p
        v += 1
    return n, v

def valuation(n, p, precision):
    n %= p**precision
    if n == 0:
        return {'value': precision, 'exact': False}
    return {'value': strip(n, p)[1], 'exact': True}

def terms(p, length, precision):
    """Terms ((1/2)_k/k!)^3; all divisions are on p-units."""
    modulus = p**precision
    unit, v = 1, 0
    for k in range(length):
        assert v >= 0
        yield (0 if 3*v >= precision else
               pow(unit, 3, modulus)*pow(p, 3*v, modulus) % modulus)
        a, va = strip(2*k+1, p)
```

```
        b, vb = strip(2*k+2, p)
        unit = unit*a*pow(b, -1, modulus) % modulus
        v += va-vb

def gaussian_trace(p):
    for a in range(1, isqrt(p)+1, 2):
        b = isqrt(p-a*a)
        if a*a+b*b == p:
            return a, b, 2*(a*a-b*b)
    raise ValueError(p)

def unit_root(p, trace, precision):
    modulus = p**precision
    u = trace % p
    assert u != 0
    for _ in range(precision):
        u = (u-(u*u-trace*u+p*p)*
             pow(2*u-trace, -1, modulus)) % modulus
    assert (u*u-trace*u+p*p) % modulus == 0
    return u

def check():
    records = []
    for p in [5, 13, 17, 29, 37, 41]:
        a, b, trace = gaussian_trace(p)
        for s in range(1, 7):
            if p**s > 100000:
                break
            precision = 3*s+3
            modulus = p**precision
            u = unit_root(p, trace, precision)
            current = list(terms(p, p**s, precision))
            previous = list(terms(p, p**(s-1), precision))
            half = sum(current[:(p**s+1)//2]) % modulus
            prevhalf = sum(previous[:(p**(s-1)+1)//2]) % modulus
            full = sum(current) % modulus
            prevfull = sum(previous) % modulus
            if s == 1:
                direct = sum(pow(comb(2*k, k), 3, modulus)*
                             pow(64, -k, modulus) for k in range(p)) % modulus
                assert full == direct
            rec = {'p': p, 's': s, 'a': a, 'b': b,
                   'precision': precision, 'modulus': modulus, 'unit_root': u,
                   'half': half, 'previous_half': prevhalf,
                   'full': full, 'previous_full': prevfull,
                   'half_defect': valuation(half-u*prevhalf, p, precision),
                   'full_defect': valuation(full-u*prevfull, p, precision),
                   'tail': valuation(full-half, p, precision),
                   'tail_defect': valuation(full-half-u*(prevfull-prevhalf),
                                            p, precision)}
            assert rec['half_defect'] == {'value': 3*s, 'exact': True}
            assert rec['full_defect'] == {'value': 3*s, 'exact': True}
            records.append(rec)
    return {'scope': 'Finite checks, not a proof of untested cases.',
            'prime_level_pairs': len(records), 'all_passed': True,
            'all_half_and_full_defects_exactly_3s': True, 'records': records}

if __name__ == '__main__':
    result = check()
    Path(__file__).with_suffix('.json').write_text(json.dumps(result, indent=2)+'\n')
    print(json.dumps({k:v for k,v in result.items() if k != 'records'}))
```

B.2 Raw formal eta-product check

Save as `check_raw_formal.py`.

```
from math import comb
import numpy as np
import json
from pathlib import Path
```

```
def mul(a,b,N,mod):
 return np.convolve(a,b)[:N+1]%mod

def eta_product(N, exponents, mod):
 a=np.zeros(N+1,dtype=np.int64);a[0]=1
 for d,e in exponents.items():
  for n in range(1,N//d+1):
   step=d*n
   b=np.zeros(N+1,dtype=np.int64)
   for k in range(N//step+1):
    if e>=0:
     if k>e: break
     coeff=(-1)**k*comb(e,k)
    else: coeff=comb(-e+k-1,k)
    b[k*step]=coeff%mod
   a=mul(a,b,N,mod)
 return a

def substitute_p(a,p,N):
 b=np.zeros(N+1,dtype=np.int64)
 b[:p*(N//p)+1:p]=a[:N//p+1]
 return b

def check(p,s,m,N,extra=1):
 mod=p**(3*s+extra)
 assert (N+1)*(mod-1)**2<2**63
 R=eta_product(N,{1:12,4:12,2:-24},mod)
 theta=eta_product(N,{2:20,1:-8,4:-8},mod)
 rr=mul(R,R,N,mod)
 u=np.zeros(N+1,dtype=np.int64);u[1:]=rr[:N]
 curr=np.zeros(N+1,dtype=np.int64);prev=curr.copy();power=curr.copy();power[0]=1
 lim=min((m*p**s-1)//2,N)
 lower=(m*p**(s-1)-1)//2
 for j in range(lim+1):
  aj=pow(comb(2*j,j),3,mod)
  curr=(curr+aj*power)%mod
  if j<=lower:prev=(prev+aj*power)%mod
  if j<lim:power=mul(power,u,N,mod)
 defect=(mul(theta,substitute_p(prev,p,N),N,mod)-mul(substitute_p(theta,p,N),curr,N,mod))%mod
 req=p**(3*s)
 bad=[(int(j),int(v)) for j,v in enumerate(defect) if v%req]
 nonzero=[int(j) for j,v in enumerate(defect) if v]
 return dict(p=p,s=s,m=m,q_even_degree=N,required_exponent=3*s,extra_precision=extra,failures=bad[:5],
     first_nonzero_at_extra_precision=nonzero[:8])

if __name__=='__main__':
 out=[]
 for p,s,m,N in [(3,1,1,80),(3,2,1,100),(3,3,1,150),(3,4,1,180),(5,1,1,100),(5,2,1,150),(5,3,1,180)
     ,(5,3,2,220),(5,3,3,260),(5,3,5,360),(7,2,1,180),(7,2,3,220),(11,2,1,200)]:
  r=check(p,s,m,N)
  out.append(r);print(json.dumps(r),flush=True)
 Path(__file__).with_suffix('.json').write_text(json.dumps(out,indent=2))
```

B.3 Supersingular half-cutoff recurrence

Save as `check_supersingular.py`.

```
"""Exact finite checks of the half-cutoff supersingular recurrence.

This program does not establish an untested prime or level.
"""
from pathlib import Path
import json
from math import comb

def strip(n, p):
    v = 0
    while n % p == 0:
        n //= p
        v += 1
```

```
    return n, v

def val(n, p, e):
    n %= p**e
    return {'value': e, 'exact': False} if not n else {'value': strip(n,p)[1], 'exact': True}

def levels(p, max_s):
    e=3*max_s+5
    mod=p**e
    ends={(p**s-1)//2:s for s in range(max_s+1)}
    out={}
    unit=1;v=0;total=0
    for k in range((p**max_s+1)//2):
        term=0 if 3*v>=e else pow(unit,3,mod)*pow(p,3*v,mod)%mod
        total=(total+term)%mod
        if k in ends:
            s=ends[k];out[s]=total
        a,va=strip(2*k+1,p);b,vb=strip(2*k+2,p)
        unit=unit*a*pow(b,-1,mod)%mod;v+=va-vb
        assert v>=0
    direct=sum(pow(comb(2*k,k),3,mod)*pow(64,-k,mod) for k in range((p+1)//2))%mod
    assert out[1]==direct
    result=[]
    for s in range(1,max_s+1):
        row={'p':p,'s':s,'precision':e,'T':out[s],'valuation_T':val(out[s],p,e)}
        if s>=2:
            defect=(out[s]-p*p*out[s-2])%mod
            row.update(defect=defect,valuation_defect=val(defect,p,e),target=3*s-1)
            row['proved_bound']=3*s-1-int(p==3)
            assert row['valuation_defect']['value']>=row['proved_bound']
        result.append(row)
    return result

if __name__=='__main__':
    rows=[]
    for p,s in [(3,10),(7,6),(11,5),(19,4),(23,4),(31,3),(43,3),(47,3)]:
        chunk=levels(p,s);rows.extend(chunk)
        print(json.dumps({'p':p,'T_values':[r['valuation_T']['value'] for r in chunk],
                          'defect_values':[r['valuation_defect']['value'] for r in chunk if r['s']>=2]}),
     flush=True)
    Path(__file__).with_suffix('.json').write_text(json.dumps({
        'scope':'Finite checks, not a proof of untested cases.',
        'recurrence_pairs':sum(r['s']>=2 for r in rows),
        'all_passed':True,'records':rows},indent=2)+'\n')
```

References

[1] H. Swisher, *On the supercongruence conjectures of van Hamme*, Research in the Mathematical Sciences **2** (2015), 18; doi:10.1186/s40687-015-0037-6; arXiv:1504.01028, Table 2, (H.3).

[2] V. J. W. Guo, *Dwork-type q-congruences through the q-Lucas theorem*, Revista de la Real Academia de Ciencias Exactas, Físicas y Naturales. Serie A. Matemáticas **119** (2025), 97; doi:10.1007/s13398-025-01762-9; arXiv:2310.15207, equations (1.9)–(1.10).

[3] F. Beukers, *Supercongruences using modular forms*, arXiv:2403.03301v3, especially the modular trace argument and Appendix C, Section 9.1.

[4] F. Beukers and M. Vlasenko, *Dwork crystals III: from excellent Frobenius lifts towards supercongruences*, arXiv:2105.14841.

[5] N. M. Katz, *p-adic properties of modular schemes and modular forms*, in Modular Functions of One Variable III, Lecture Notes in Mathematics **350** (1973), 69–190; author's scan. See Proposition 2.7.1, Sections 3.1–3.4, and Appendix A2, in particular equations (A2.2.1) and (A2.2.7), printed pp. 175–177.

[6] P. Berthelot and W. Messing, *Théorie de Dieudonné cristalline I*, Astérisque **63** (1979), 17–37; original text. See Section 2.1 for connections and lift comparison, Theorem 2 and the Frobenius–Verschiebung constructions in Section 2, and Section 3, equation (3.1), for the abelian-scheme cohomology and Hodge filtration.

[7] R. Shirai, *q-deformation with* (φ, Γ) *structure of the de Rham cohomology of the Legendre family of elliptic curves*, arXiv:2006.12310. The introduction records the Legendre basis, crystalline comparison, cup product, and the sign $(-1)^{(p-1)/2}$ in the unit-root normalization; Definition 3.1 records Frobenius–Hodge divisibility.

[8] The Stacks Project Authors, *The Stacks Project*, Proposition 60.17.4 (07JH), crystalline Taylor comparison; Remark 60.24.14 (07MZ), integral completed crystalline–de Rham comparison; and Theorem 60.26.4 (07N5), the canonical integral Frobenius map. Accessed 10 November 2026.

[9] G. Ligozat, *Courbes modulaires de genre 1*, Mémoires de la Société Mathématique de France **43** (1975), 5–80, Proposition 3.2.8; original text.

[10] NIST Digital Library of Mathematical Functions, Section 20.7, theta identities and transformations. Accessed 10 November 2026.